

Template For Abstracts For The 31st Annual Conference on Geology of Long Island and Metropolitan New York.

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Title: = Microscopic Characterization of Particulate Contaminants in Snowfall at a Suburban Long Island Site

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Abstract:

Snowfall can act as an effective sampler of atmospheric particulates, capturing a wide range of contaminants that are later released into the environment during snowmelt. This study provides a preliminary microscopic characterization of particulate contaminants in urban snowfall, establishing a methodological framework for future geochemical analysis of heavy metal deposition in winter precipitation.

Snow samples were collected from a suburban site on Long Island, specifically the rooftop of Gittleson Hall at Hofstra University in Hempstead, New York, during two snowfall events in 2026. Controlled

collection procedures were used to minimize contamination, followed by melting, filtration, and laboratory preparation.

Collected particulates were analyzed using optical microscopy and scanning electron microscopy (SEM) to examine morphology, size distribution, and aggregation patterns. Observations revealed a heterogeneous mixture of contaminants, including fine particles, fibrous materials, and irregularly shaped aggregates. SEM imaging identified clustered and porous structures, suggesting the presence of complex, multi-source atmospheric particulates likely influenced by urban and suburban activities.

Although elemental composition was not determined in this phase of the study, the observed diversity of particles indicates multiple potential sources, including anthropogenic emissions (such as road salt) and natural debris. These findings support the role of snowfall as a temporary reservoir for atmospheric contaminants and highlight its importance in transporting particulate matter into terrestrial and aquatic systems during melt events.

Introduction: This study investigates particulate contaminants captured in snowfall collected from the roof of Gittleston Hall at Hofstra University in Hempstead, New York, serving as a pilot site to understand urban and suburban atmospheric deposition processes. Snowfall and snowmelt in New York City and Long Island are capable of accumulating particulate contaminants, including potentially heavy metals and urban pollutants that leach from urban pollution, which poses significant risks to water quality and public health. Snow in a suburban site on Long Island (Hofstra University, Hempstead, NY) acts like a “toxic sponge,” absorbing heavy metals such as copper, zinc, and lead along with other urban pollutants before releasing a concentrated “toxic cocktail” during melting. This release of contaminants can harm aquatic ecosystems and pose public-health risks as heavy-metal-laden runoff enters streams, rivers, and coastal waters. This study investigates the concentration of these contaminants through the collection of snow samples, melting them, and analyzing their morphology and distribution using optical microscopy and SEM, with future work aimed at determining elemental composition.

Methodology:

Include a photo of the sampling time

Snow samples were collected from the top of Giddleston Hall at Hofstra University from 2 separate snowfall events. The first set of samples was collected on 1-28-26, and the second set of samples was collected on 2-25-26. Sampling was conducted on the rooftop of Gittleston Hall at Hofstra University (Hempstead, NY) to minimize ground-level contamination. In order to reduce the chance for sample contamination from sources like soil, debris, and surface water run-off, samples were collected 3 inches above the ground rather than at ground level. In addition, the bottom 3 inches from the top of the snowfall were avoided in order to reduce sample contamination from airborne sources after the snowfall. This method allowed for the analysis of the middle part of the snowfall samples, which was most representative of any pollution that was accumulated during the snowfall event. Clean collection equipment was used during the collection procedure in order to maintain consistency among all sample

collection sites. Once collected, the snow samples were transported for melting, filtration, and analyze the morphology, size distribution, and spatial clustering of particulate contaminants



Figure 1: Snow sampling procedure on the rooftop of Gittleston Hall, Hofstra University (Hempstead, NY) during the January 28, 2026 snowfall event. Samples were collected from the undersnowpack to minimize contamination from surface runoff and post-depositional atmospheric inputs. ~~Procedure for collecting~~
~~Snowfall from Giddleston Hall on 1-28-26~~

During the initial phase of the project, a preliminary filtration system was constructed by modifying a large water container and integrating it into a laboratory sink setup. This system was used to process water samples and refine collection and filtration techniques. Building on this approach, the study was designed to collect snowfall and snowmelt samples from a typical suburban site near Hempstead, NY how density and land-use characteristics influence pollutant deposition in winter precipitation.

Following method development, collected samples were prepared and analyzed using optical microscopy and scanning electron microscopy (SEM) to characterize particulate morphology and distribution. Although the scope of analysis was constrained to microscopic characterization, these observations provided valuable insight into the physical properties and diversity of particulate matter present in urban snowfall. This iterative approach allowed for refinement of sampling and analytical methods while establishing a foundation for continued investigation. Future work will expand on these methods to further evaluate spatial variability and contaminant sources across urban and suburban environments.



Fugue 2: Water jug diagram
with a filter below the opening

For scanning electron microscopy (SEM) preparation, a small volume of clean water was carefully applied to the filter surface to mobilize and direct the sediment toward the middle region of the filter for the purposes of the SEM. Although complete consolidation of all particulates was not achievable due to some particles sticking in the filter, the majority of the sediment was successfully concentrated into a more uniform area to facilitate efficient SEM imaging and analysis. The filter was then allowed to dry completely. Once dried, the central portion containing the concentrated particulates was trimmed to an appropriate size and mounted onto an SEM specimen stub using conductive adhesive to reduce charging effects. The prepared specimen was introduced into the SEM chamber, vacuum conditions were established, and imaging parameters were optimized through the instrument's control software prior to conducting high-resolution morphological characterization of the collected contaminants.

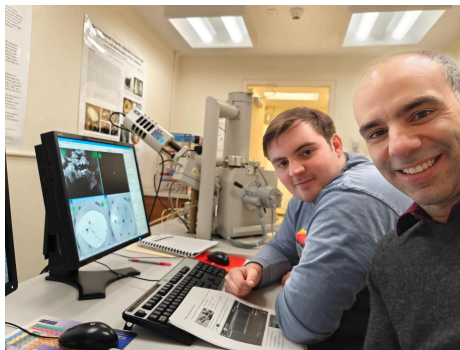


Figure 3: SEM Procedure

Results:

Optical Microscopy Observations

From your images:

- **Presence of:**
 - **Dark aggregated particles**

- Fibrous material (likely organic or textile)
- Granular debris (possibly mineral dust)
- Particle sizes vary from:
 - Fine (<100 μm)
 - Coarse aggregates (>500 μm)

👉 Interpretation:

- Indicates multiple contamination sources:
 - Atmospheric deposition
 - Urban dust
 - Organic debris
-

B. SEM Observations

From your SEM image:

- Irregular, porous, clustered structures
- Non-spherical morphology (important!)
- Aggregation of fine particles into larger masses

👉 Interpretation:

- Suggests:
 - Secondary aggregation during melting/filtration
 - Possible soot-like or combustion-derived particles
 - Not typical of pure mineral grains (which are more angular/crystalline)
-

🔑 Important language:

DO NOT say:

✗ “heavy metals were observed”

Instead say:

✓ “particles consistent with urban atmospheric contamination were observed”

Despite occasional struggles with the setup of the filter experiment, we were able to garner sufficient results from the experiment. From the in-lab procedure, we detected dark aggregated particles, fibrous material, and granular debris through the filter experiment. The particles varied in size from fine

to coarse aggregates. Based on the results through the filter, there was an indication of multiple contamination sources, including atmospheric deposition, urban dust, and organic debris.



Figure 4: Filtered particles from snowfall results

Under the microscope, it was observed that the snow samples contained different kinds of particulate impurities. The more prominent impurities observed were dark aggregated particles, which might be organic or textile in nature. The size of these impurities ranges from smaller impurities of sizes less than 100 μm to larger impurities larger than 500 μm . The presence of granular impurities, which might be mineral dust, was also observed. The presence of salt particles and asphalt was also observed, which might be due to anthropogenic sources

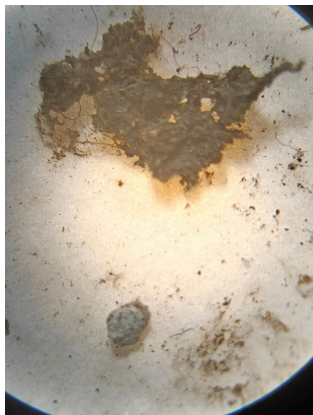


Figure 5: Filtered particles under the microscope

Despite some minor challenges during filter preparation, sufficient particulate material was collected for SEM analysis. SEM imaging revealed irregular, non-spherical particles forming porous, clustered aggregates with diffuse and filamentous surface textures. The particles varied in size from fine to coarse structures, with no well-defined crystalline features typical of primary mineral grains. The observed morphology indicates a mixture of particle types and suggests the presence of aggregated material formed through secondary processes. These characteristics are consistent with particles

associated with urban atmospheric contamination. However, further SEM imaging at higher magnifications and energy-dispersive spectroscopy (EDS) analysis will be necessary to more accurately determine the composition and origin of the collected material.

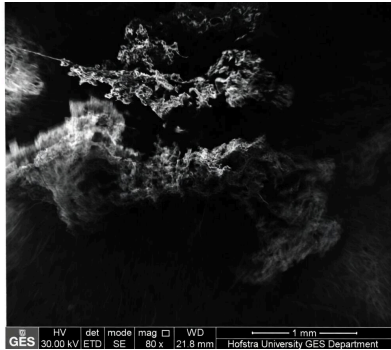


Figure 6: SEM Result

Discussion:

Key points to include:

The rooftop sampling location provides a controlled environment that minimizes contamination from direct ground contact, allowing for a clearer representation of atmospheric particulate deposition.

✓ Urban Signature

- Mixed particle types → urban heterogeneity
- Fibers → anthropogenic (clothing, construction, indoor/outdoor sources)
- Fine dark particles → possible combustion/traffic origin

✓ Snow as a Transport Medium

Link back to your original idea:

Snow acts as a temporary reservoir and transport medium for urban particulates

✓ Environmental Implication

Even without EDS:

- These particles will enter:
 - runoff

- **soil**
- **waterways**

Future Work (VERY IMPORTANT):

Here is where you **strategically introduce heavy metals**:

You can write:

“Future work will include elemental analysis using SEM-EDS or PXRF to quantify heavy metal concentrations and confirm the chemical composition of observed particulates.”

Also add:

- Spatial comparison (urban vs suburban)
- Temporal variation (multiple snow events)
- Particle size distribution analysis

Also somewhere at the end:

While this study focuses on a single suburban site, the methodology can be extended to broader spatial analyses across New York City and Long Island.

Figure 4 shows the filter after melting the snow, which captures the dark aggregated particles, fibrous material, and granular debris we consistently observed in our samples. This visual indicates that suburban snowfall in Hempstead carries a mix of particulate matter collected during the snowfall event. The collected material serves as a reflection of multiple contamination sources, including atmospheric deposition and urban dust, as noted in our results. By documenting these particles at the filtration stage, the figure sets up the microscopic analysis that follows and reinforces snow’s role as a temporary reservoir for urban contaminants.

In the optical microscopy image, a mixed composition of particulate material is visible, which was collected from the snowfall sample. The particulate matter includes dark-collected aggregated material, fibers, and granular material. The particulate matter is of different shapes and sizes, as some of the material is finer in texture and other material is present in clumps of bigger sizes. The presence of dark particulate matter might be due to contamination of the atmosphere due to urban areas, and the fibers might be due to organic or textile material contamination. The granular material might be due to dust or salt present on the road due to snowfall. The presence of a variety of particulate material in a single image indicates that the snow might be collecting contaminants from various natural and human-related sources. The image also supports the idea of snowfall acting as a reservoir for particulate matter pollution.

The SEM image shows irregular, non-spherical particles forming porous, clustered aggregates with diffuse and filamentous surface textures, indicating a heterogeneous structure. These features suggest the particles are agglomerates formed through secondary processes such as atmospheric interactions rather than primary crystalline mineral grains. The variation in particle size and lack of defined crystal faces further support a mixed origin, consistent with urban atmospheric particulate matter. The morphology also implies a high surface area, which may enhance the interaction of particles with surrounding environmental compounds. Future research will include higher magnification SEM imaging and energy-dispersive spectroscopy (EDS) to better characterize particle structure and determine elemental composition. While this study focuses on a single suburban site, the methodology can be extended to broader spatial analyses across New York City and Long Island.

Conclusions:

This study represents a pilot investigation of snowfall-mediated particulate deposition at a suburban Long Island site, providing a foundation for future multi-site and geochemical analyses

This study represents a pilot investigation of snowfall-mediated particulate deposition at a suburban Long Island site, providing a foundation for future multi-site and geochemical analyses. Our procedures clearly indicate that suburban snowfall collected on the Gittelsohn Hall rooftop at Hofstra University serves as a temporary reservoir for mixed urban particulates, with our optical and SEM analyses revealing dark aggregated particles, fibrous material, granular debris, and irregular clustered structures that point to diverse atmospheric contamination sources. These results suggest that snowfall can carry particles in suburban areas and move them when the snow melts. This shows why it is important to keep studying how these particles may affect nearby soil and water. Future work will involve higher magnification SEM imaging to further resolve particle morphology, along with energy-dispersive spectroscopy (EDS) to determine elemental composition. These analyses will provide a more detailed understanding of particle structure and composition.

To all teams: This is a suggested statement please revise with your names and work accordingly

Credit Authorship Contribution Statement

Badger, T.: editing, literature, Rstudio coding, Figure 4, Figure 5, writing- Results, Discussion; Parag, A.: Literature, editing, writing- Global Mapper, Google Earth, Abstract, Results- figure 1, Methods, Discussion; Pelletier, A.: references, literature, editing, writing - Introduction, Discussion; DeRocchis, S.: figures, editing, USGS Graphs, Global Mapper, writing - Methods, Discussion; Marsellos, A.E.: supervision, Rstudio coding, guidance, editing; Tsakiri, K.G.: Rstudio coding.